

6

CHAPTER

Steady Currents

Learning Objectives

After reading this chapter, you should be able to:

- ▶ Define current and current density and develop the fundamental continuity equation.
- ▶ State and explain Kirchhoff's laws for the distribution of currents in a network of conductors.
- ▶ Apply Kirchhoff's laws to deduce the condition of balance of a Wheatstone network.
- ▶ Explain with necessary theory how the Carey Foster's bridge may be used to compare two nearly equal resistances.
- ▶ Explain the theory of a potentiometer. Explain how it is used for measuring current, voltage and resistance.
- ▶ State and explain Thevenin's and Norton's theorems.

6.1 CURRENT AND CURRENT DENSITY

Electric current. In an isolated metallic conductor, the free electrons present in it are in random motion like the molecules of a gas. So the net rate at which electrons pass through any hypothetical plane is zero. If the ends of the conductor are connected to a battery, an electric field $\mathbf{E}$ will be set up at every point within the conductor. This field acts on the electrons and gives them a resultant motion in the direction $-\mathbf{E}$. If a charge q passes through any cross-section of the conductor in time t , then the current i is defined by

$$i = q/t$$

When q is in coulomb and t in second, then i is in ampere. If the rate of flow of charge with time is not constant, the current varies with time. Then at any instant t current is defined as

$$i = dq/dt$$

So *current is defined as the net charge flowing across the area per unit time*. Current is a scalar quantity. Conventionally, the direction of electric current is taken along the direction of motion of positive charges. When the current is caused by the motion of electrons, the direction of current is opposite to the direction of electron flow.

Current density. Current is a macroscopic quantity. The related microscopic quantity is the *current density $\mathbf{J}$* .

The current density at any point is defined as the quantity of charge passing per second through a unit area taken perpendicular to the direction of the flow of charge at that point.

It is a vector quantity and is characteristic of the given point inside a conductor. If the current is flowing uniformly across a conductor of cross-sectional area A , the *magnitude* of the current density for all points on that cross section is given by

$$J = \frac{q/t}{A} = \frac{i}{A}$$

The unit of current density is A/m^2 .

Current i is related to the *current density $\mathbf{J}$* by

$$i = \int \mathbf{J} \cdot d\mathbf{S}$$

where $d\mathbf{S}$ is an element of area and the integral is taken over any surface cutting across the conductor.

From this equation we may define *the current i as the flux of the current density vector $\mathbf{J}$ through a given area*.

6.2 EXPRESSION FOR CURRENT DENSITY

Consider a wire of length L and area of cross section A in which there is an electric field $\mathbf{E}$ towards the left (Fig. 6.1). Let n be the number of free electrons per metre^3 . The free electrons move towards the right with a constant drift velocity v_d .

Number of conduction electrons in the wire $= n(AL)$.

Let e be the charge on each electron.

Magnitude of charge q , passing out of the wire in time t is

$$q = (nAL)e$$

where t is given by

$$t = L/v_d$$

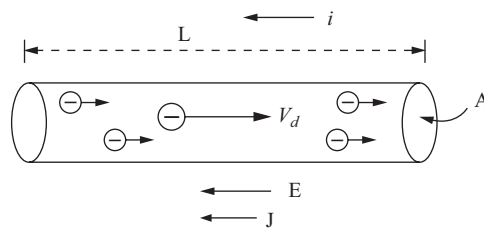


FIG. 6.1

The current in the wire is

$$i = \frac{q}{t} = \frac{nALe}{L/v_d} = nAev_d$$

But i/A is the current density J . Therefore

$$J = nev_d$$

In vector form,

$$\mathbf{J} = (ne) \mathbf{v}_d \quad \dots(1)$$

(ne) is the volume charge density ρ of the current carriers. For positive carriers ne is positive and Eq. (1) predicts that $\mathbf{J}$ and $\mathbf{v}_d$ point in the same direction. For electrons $\mathbf{J}$ and $\mathbf{v}_d$ point in opposite directions.

6.3 EQUATION OF CONTINUITY

Consider a closed surface S enclosing a region of volume V . Let the surface enclose some charge within it. The current i flowing *out* through the surface S is given by

$$i = \oint_S \mathbf{J} \cdot d\mathbf{S} \quad \dots(i)$$

where $\mathbf{J}$ is the current-density vector over a cross-section $d\mathbf{S}$.

Let ρ be the *charge density* (charge residing per unit volume) of the charge enclosed within the surface at any instant. Then the total charge enclosed at any instant is given by

$$q = \int_V \rho \, dV$$

The current i is the rate of decrease of charge inside the volume, so that

$$i = -\frac{dq}{dt} = -\frac{d}{dt} \int_V \rho \, dV = -\int_V \frac{d\rho}{dt} \, dV \quad \dots(ii)$$

According to Gauss divergence theorem, $\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{J}) \, dV$

$$\therefore \oint_S \mathbf{J} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{J}) \, dV = -\int_V \frac{d\rho}{dt} \, dV$$

or

$$\int_V \left\{ (\nabla \cdot \mathbf{J}) + \frac{d\rho}{dt} \right\} \, dV = 0$$

This integral must be zero for any arbitrary volume. So

$$\nabla \cdot \mathbf{J} + \frac{d\rho}{dt} = 0 \quad \dots(iii)$$

This equation is called the *equation of continuity*. It simply expresses the fact that the charge is conserved.

In most practical situations of interest, the charge density is constant in time at each point. This means $d\rho/dt$ is zero at each point. This is called a *steady state*. Hence for *steady currents*

$$\nabla \cdot \mathbf{J} = 0 \quad \dots(iv)$$

6.4 OHM'S LAW AND ELECTRICAL CONDUCTIVITY

It is found experimentally that in a metallic conductor at constant temperature, the current density $\mathbf{J}$ is linearly proportional to the electric field $\mathbf{E}$. Thus

$$\mathbf{J} = \sigma \mathbf{E} \quad \dots(1)$$

The equation $\mathbf{J} = \sigma \mathbf{E}$ is general vector statement of Ohm's law. Here σ is a constant called the *conductivity* of the conductor. The unit of conductivity is mho/metre. The reciprocal of the conductivity is called the *resistivity* ρ . In terms of ρ Eq. (1) can be written as

$$\mathbf{E} = \rho \mathbf{J} \quad \dots(2)$$

In Fig. 6.2 the electric field along the rod is in the direction MN and its value is $E = V/L$ everywhere. Here V is the total potential drop from M to N . Thus

$$J = \sigma(V/L)$$

The total current $= i = JA = \sigma VA/L$.

Here A is the cross-sectional area of the rod. This leads to

$$\frac{V}{i} = \frac{L}{\sigma A} = \frac{\rho L}{A} \quad \dots(3)$$

The ratio V/i is called the *resistance* R of the rod. $R = \rho L/A$.

Eq. (3) becomes $V = iR$

This is known as Ohm's law.

Experiments show that generally ρ and therefore R increase with temperature according to

$$R_t = R_0 (1 + \alpha t)$$

where α is called the *temperature coefficient of resistivity*, and t is the temperature in degrees centigrade.

The *power* dissipated in a resistance R carrying a current i is

$$P = i^2 R = iV = V^2/R$$

Example. A copper wire of diameter 0.5 mm and length 20 m is connected across a battery of emf 1.5 V and internal resistance 1.25 Ω . Calculate the current density in the wire and the drift velocity v_d , assuming one conduction electron per atom of copper. What is the heat dissipated per metre of the wire?

Solution. The total resistance of the wire is

$$R = \frac{\rho L}{A} = \frac{(1.7 \times 10^{-8}) \times 20}{\pi(0.25 \times 10^{-3})^2} = 1.732 \Omega$$

$$\text{Current} = i = \frac{V}{R_{\text{total}}} = \frac{1.5}{(1.732 + 1.25)} = 0.5 \text{ A}$$

Resistance of unit length of wire $= 1.732/20 = 0.0866 \Omega/\text{m}$

Power dissipated per metre $= i^2 R = (0.5)^2 (0.0866) = 0.0216 \text{ W/m}$

The number of copper atoms per m^3 is

$$\begin{aligned} n &= \frac{\text{Avogadro no.} \times \text{density}}{\text{atomic weight}} \\ &= \frac{(6.025 \times 10^{26}) (8.89 \times 10^3)}{63.54} = 8.43 \times 10^{28} \end{aligned}$$

Georg Simon Ohm (1787–1854)

Ohm was led to his law by an analogy between the conduction of heat: the electric field is analogous to the temperature gradient, and the electric current is analogous to the heat flow.

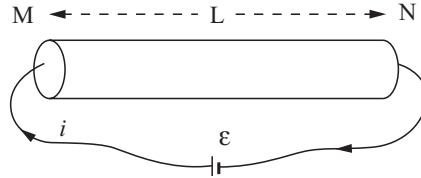


FIG. 6.2

A battery of emf ϵ can maintain an electric field within a conducting rod.

$$\text{Current density, } J = \frac{i}{A} = \frac{0.5}{\pi (0.25 \times 10^{-3})^2} = 2.546 \times 10^6 \text{ A m}^{-2}$$

$$v_d = \frac{J}{ne} = \frac{2.546 \times 10^6}{(8.43 \times 10^{28})(1.6 \times 10^{-19})} = 0.1888 \times 10^{-3} \text{ ms}^{-1}$$

6.5 DRUDE-LORENTZ THEORY OF ELECTRICAL CONDUCTION

The basic assumptions of Lorentz-Drude theory are:

1. There are a large number of free electrons in a metal. These electrons are free to move about the whole volume of the metal like the molecules of a perfect gas in a container.
2. The free electrons make collisions from time to time with fixed positive ions in the lattice and also among themselves. We may neglect the collisions between electrons in comparison with the collisions between electrons and ion cores.
3. In the absence of electric field, the random motion of free electrons is equally probable in all directions so that current density vector is zero.

4. When an external electric field is applied, the electrons drift slowly with some average velocity, known as *average drift velocity*, in the direction opposite to that of electric field. The drift velocity of free electrons is superimposed over their random velocity. In Fig. 6.3, the continuous solid lines show a possible random path followed by an electron in the absence of an applied field. The dashed lines show the electron path in the presence of the electric field $\mathbf{E}$. Note the steady drift in the direction of $-\mathbf{E}$. The drift velocity is much smaller in magnitude than average random velocity of free electrons.

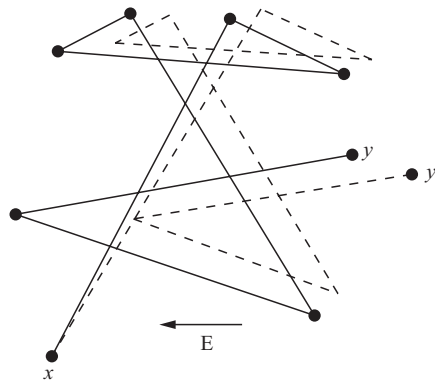


FIG. 6.3

5. The average distance traversed by a free electron between two successive collisions with the positive ions is called *mean free path* and is denoted by λ .

If an electron of mass m and charge e is placed in an electric field E , it experiences an acceleration a given by

$$a = eE/m$$

Let τ be the average time-interval between two successive collisions of an electron with positive ions.

Increase in drift velocity in time $\tau = a\tau = eE \tau/m$

This drift velocity is reduced to zero at the next collision and again builds up to the value $eE\tau/m$ before suffering another collision and so on.

$$\therefore \left. \begin{array}{l} \text{the average drift velocity} \\ \text{through the conductor} \end{array} \right\} = v_d = \frac{eE\tau}{2m}$$

Let n be the number of conduction electrons per unit volume.

$$\text{Current density} = J = nev_d = \frac{ne^2 E \tau}{2m}$$

But $J = \sigma E$ where σ is the electrical conductivity of the metal.

$$\therefore \sigma = \frac{J}{E} = \frac{ne^2 \tau}{2m}.$$

This is the expression for the electrical conductivity of a metal.

(Note. Some authors write the above equation without the factor $1/2$.)

$$\text{Resistivity} = \rho = 1/\sigma = 2m/ne^2\tau$$

Let λ be the mean free path and v the r.m.s. thermal velocity of electrons. Then $\tau = \lambda/v$

$$\text{Electrical conductivity, } \sigma = \frac{ne^2\lambda}{2mv}$$

$$\text{The kinetic energy of a free electron, } \frac{1}{2}mv^2 = \frac{3}{2}kT.$$

$$\therefore \sigma = \frac{ne^2 \lambda v}{2mv^2} = \frac{ne^2 \lambda v}{6kT}$$

Deduction of Ohm's Law from Drude-Lorentz Theory

$$\text{From Drude Lorentz theory, } J = \sigma E = \frac{ne^2\lambda}{2mv} E$$

At a particular temperature, λ and v are constant.

$$\therefore \mathbf{J} \propto \mathbf{E}$$

Let i be the current flowing through the conductor of length l and area of cross-section A and V be the potential difference across it. Then

$$J = \frac{i}{A} \text{ and } E = \frac{V}{l}$$

The above relation becomes

$$\frac{i}{A} \propto \frac{V}{l}$$

For a given conductor, l and A are constant at a given temperature.

$$\therefore i \propto V$$

i.e., at a given temperature, the current flowing through a conductor is proportional to the potential difference between its two ends, which is Ohm's law.

Variation of Electrical Conductivity with Temperature

The electrical conductivity of metals is given by

$$\sigma = \frac{ne^2\tau}{2m}$$

According to the kinetic theory of gases (here we are dealing with electron-gas) the time-interval τ between two collisions is inversely proportional to the square-root of the absolute temperature:

$$\tau \propto \frac{1}{v_{rms}} \propto \frac{1}{\sqrt{T}}.$$

Therefore, the conductivity σ must decrease with increase in temperature. Experimentally, however, σ is inversely proportional more nearly to the first power of the absolute temperature.

Wiedemann-Franz Law. By an analysis similar to that used for electrical conductivity, it may be shown that the thermal conductivity K is given by

$$K = \left(\frac{3nk^2\tau}{2m} \right) T,$$

where k is Boltzmann's constant and T is the absolute temperature.

$$\therefore \frac{K}{\sigma} = 3 \left(\frac{k}{e} \right)^2 T.$$

This relation is known as the *Wiedemann-Franz law*.

Statement. The ratio of thermal conductivity to electrical conductivity is a constant for all metals and is proportional to the absolute temperature.

6.6 KIRCHHOFF'S LAWS

Law 1. In any network of conductors, the algebraic sum of the currents meeting at any point is zero, i.e., $\Sigma I = 0$.

In the above equation, current flowing into a junction is regarded as positive, while the current flowing out of the junction is regarded as negative. In Fig. 6.4, O is a junction.

According to sign convention, I_1, I_2, I_5 are positive and I_3, I_4, I_6 are negative. Then,

$$I_1 + I_2 - I_3 - I_4 + I_5 - I_6 = 0$$

$$\text{i.e.,} \quad I_1 + I_2 + I_5 = I_3 + I_4 + I_6$$

Thus the sum of currents entering the point O is equal to the sum of currents leaving it. In other words Kirchhoff's first law states, "When steady current flows in an electric circuit, there is no accumulation of charge at any point or junction."

Law 2. The algebraic sum of the products of the current and resistance in any closed loop of a circuit is equal to the algebraic sum of electromotive forces (e.m.f.'s) acting in that loop, i.e., $\Sigma IR = \Sigma E$.

Sign convention:

- (i) A product of current and resistance is taken as positive when we traverse in the direction of the current.
- (ii) The emf is taken as positive when we traverse from the negative to the positive electrode of the cell through the electrolyte.

Let i_1 and i_2 be the currents through R_1 and R_2 (Fig. 6.5). Applying first law to the junction A , the current through R is $i_1 + i_2$. Then applying the second law for the closed loop $ZAYZ$

$$i_1 R_1 + (i_1 + i_2) R = E_1$$

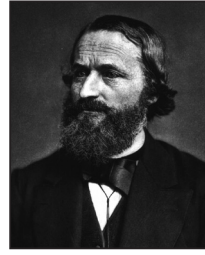
Similarly, for the closed loop $XAYX$,

$$i_2 R_2 + (i_1 + i_2) R = E_2$$

For the closed loop $ZAXYZ$,

$$i_1 R_1 - i_2 R_2 = E_1 - E_2$$

These 'Loop equations' enable us to obtain the values of currents in different parts of the circuit in terms of R, R_1, R_2, E_1 and E_2 .



Gustav Robert Kirchhoff
(1824–1887)

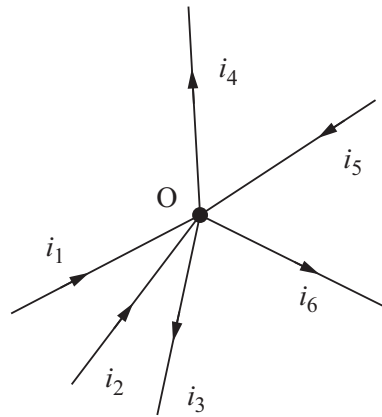


FIG. 6.4

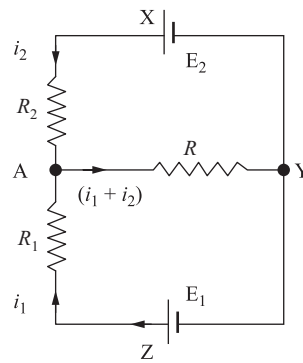


FIG. 6.5

Application of Kirchhoff's Laws to Wheatstone's Network

Four resistances P , Q , S and R form a closed network $ABCD$ (Fig. 6.6). A cell of emf E is connected between A and C . A galvanometer of resistance G is connected between B and D .

Let I be the current along EA , I_1 along AB and I_g along BD . By Kirchhoff's first law, the current along the various branches will be as shown.

We can obtain an expression for the current through the galvanometer i_g by applying Kirchhoff's voltage law (KVL).

Applying KVL to the closed mesh $ABDA$,

$$I_1 P + I_g G - (I - I_1) R = 0$$

$$\text{or} \quad (P + R) I_1 + G I_g = I R \quad \dots(1)$$

Applying KVL to the closed mesh $BCDB$,

$$(I_1 - I_g) Q - (I - I_1 + I_g) S - I_g G = 0$$

$$\text{or} \quad (Q + S) I_1 - (Q + S + G) I_g = I S \quad \dots(2)$$

Multiplying Eq. (1) by $(Q + S)$,

$$(P + R) (Q + S) I_1 + G (Q + S) I_g = R (Q + S) I \quad \dots(3)$$

Multiplying Eq. (2) by $(P + R)$,

$$(P + R) (Q + S) I_1 - (P + R) (Q + S + G) I_g = S (P + R) I \quad \dots(4)$$

Subtracting Eq. (4) from Eq. (3),

$$[G (Q + S) + (P + R) (Q + S + G)] I_g = [R (Q + S) - S (P + R)] I$$

$$\therefore I_g = \frac{(QR - PS) I}{[G (P + Q + R + S) + (P + R) (Q + S)]} \quad \dots(5)$$

Condition for balance is $i_g = 0$ i.e., $(QR - PS) I = 0$. Since I has a finite value, $QR - PS = 0$.

Thus the condition for the balance of the bridge is

$$\frac{P}{Q} = \frac{R}{S} \quad \dots(6)$$

Sensitivity of Wheatstone's Bridge

Fig. 6.7 represents a Wheatstone's bridge $ABCD$. The unknown resistance r is connected in the arm CD . Let the resistances in the arms AB , BC and AD be nmr_1 , mr_1 and nr_1 , respectively. Let G be the galvanometer resistance. Let I be the current through the unknown resistance r , I_g the current through the galvanometer and I_1 the current through resistance nmr_1 . Then, by Kirchhoff's I law, the currents through mr_1 and nr_1 are respectively $(I_1 + I_g)$ and $(I + I_g)$.

When $r = r_1$ the bridge is balanced and $I_g = 0$. Therefore, the difference $(r - r_1)$ is a measure of the *want of balance*. That is, the greater the difference $(r - r_1)$, the far is the bridge from balance. The bridge will be most sensitive, if the galvanometer current I_g is large, even when $(r - r_1)$ is extremely small.

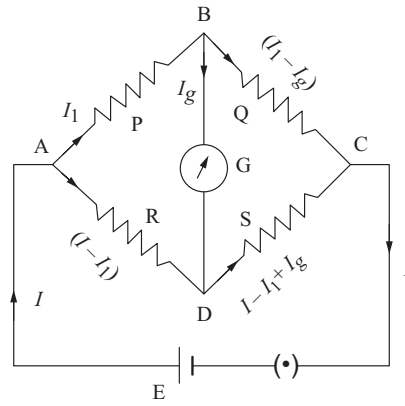


FIG. 6.6

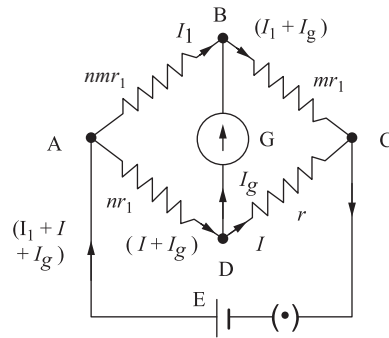


FIG. 6.7

Applying KVL to the loops $ABDA$ and $BCDB$ respectively,

$$nmr_1 I_1 - GI_g - nr_1 (I + I_g) = 0 \quad \dots(1)$$

$$mr_1 (I_1 + I_g) - rI + GI_g = 0 \quad \dots(2)$$

Multiplying Eq. (2) by n , we get

$$nmr_1 I_1 + nmr_1 I_g - nrI + nGI_g = 0 \quad \dots(3)$$

Subtracting Eq. (1) from Eq. (3), we get

$$G(1+n)I_g + nr_1(1+m)I_g - n(r-r_1)I = 0$$

$$\text{or} \quad \frac{I_g}{I} = \frac{n(r-r_1)}{G(1+n) + nr_1(1+m)}$$

$$\text{or} \quad \frac{I_g}{I} = \frac{r-r_1}{(1+1/n)G + r_1(1+m)} \quad \dots(4)$$

For a given want of balance $(r-r_1)$, the value of I_g will be large when,

(i) I , the current through the unknown resistance is large.

(ii) G , the galvanometer resistance is small.

(iii) n , the ratio $\frac{P}{Q} = \frac{nmr_1}{mr_1}$ is large.

(iv) m , the ratio $\frac{P}{R} = \frac{nmr_1}{nr_1}$ is small.

The current I cannot be increased indefinitely because this will heat up the resistance coils and change their resistances. The sensitivity of the galvanometer itself is decreased if the resistance of the galvanometer G is too low.

Therefore, we have to increase the sensitivity by increasing n and decreasing m . In the ideal case, if $n = \infty$ and $m = 0$,

$$\left(\frac{I_g}{I}\right)_{\max} = \frac{(r-r_1)}{(G+r_1)}$$

But this maximum sensitivity can never be realised in practice, because when $n = \infty$, the resistances in AB and AD are infinite. If $m = 0$, the points A and C are short circuited.

If $m = n = 1$,

$$\frac{I_g}{I} = \frac{(r-r_1)}{2(G+r_1)}$$

This value is still half of the value in the ideal case. In practice, this is the condition for maximum sensitivity. However, for good sensitiveness of the bridge, $n > 1$ and $m < 1$. Thus the Callendar rule is: *The sensitivity of the bridge will be higher if the resistance in series with the unknown resistance is greater than the resistance connected in parallel to it.*

The positions of the galvanometer and the battery can be interchanged but these two arrangements are not equally sensitive. Maxwell gave the following rule regarding the best arrangement of these.

Out of the battery and the galvanometer, the one having the higher resistance should be connected between the junction of the two highest resistances and the junction of the two lowest resistances.

Example 1. A battery of e.m.f 6 volts and internal resistance 5 ohm is joined in parallel with another of e.m.f. 10 volts and internal resistance 1 ohm and the combination sends a current through an external resistance 12 ohms. Calculate the current through each battery.

Solution. Applying Kirchhoff's II law to the mesh $ABFEA$ (Fig. 6.8),

$$I_1 \times 5 + (I_1 + I_2) \times 12 = 6$$

or $17I_1 + 12I_2 = 6 \quad \dots(1)$

Applying Kirchhoff's II law to the mesh $CDEFC$

$$I_2 \times 1 + (I_1 + I_2) \times 12 = 10$$

or $12I_1 + 13I_2 = 10 \quad \dots(2)$

Solving (1) and (2), we get

$$I_1 = -\frac{6}{11} \text{ A}, I_2 = \frac{14}{11} \text{ A}.$$

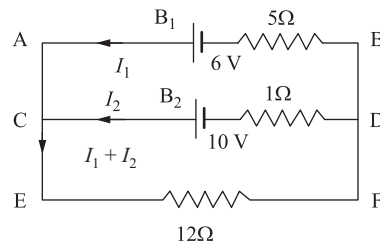


FIG. 6.8

Example 2. Twelve conductors each of resistance 3 ohm are connected to form a skeleton cube. A battery of e.m.f. 6 V and internal resistance 0.5 ohm is connected between two diagonally opposite corners of the cube. Find the equivalent resistance of the cube, and the current supplied by the battery.

Solution. Let I be the current supplied by the battery. The current I enters the system at A and leaves it at G (Fig. 6.9). The distribution of current in the various branches is as shown in Fig. 6.9.

Applying KVL to the closed loop $ABFGA$,

$$\frac{I}{3} \times 3 + \frac{I}{6} \times 3 + \frac{I}{3} \times 3 + 0.5I = 6$$

$$I + 0.5I + I + 0.5I = 6 \quad \text{i.e., } 3I = 6$$

$\therefore I = 2 \text{ A}$

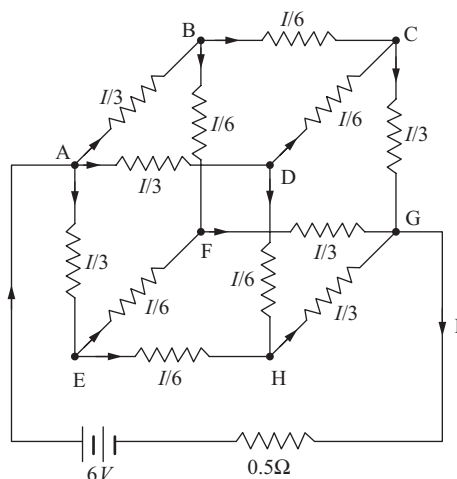


FIG. 6.9

Let X be the equivalent resistance of the frame work. Then,

$\therefore (X + 0.5) I = 6$

Substituting for I , $2X + 1 = 6$ or $2X = 5$

$\therefore X = 2.5 \Omega$

Example 3. The resistance of the four arms of a Wheatstone's net work are 5, 5, 5 and 5.2 ohms. The resistance of the galvanometer is 40 ohms and the battery supplies a current of 0.2 Amp. Calculate the current through the galvanometer.

Solution. Applying Kirchhoff's second law to the closed network $ABDA$ (Fig. 6.10),

$$5I + 40I_g - 5[0.2 - I] = 0$$

$$\text{or} \quad 10I + 40I_g = 1 \quad \dots(1)$$

For the network $BDCB$,

$$40I_g + 5.2(0.2 - I + I_g) - 5(I - I_g) = 0$$

$$50.2I_g - 10.2I = -1.04 \quad \dots(2)$$

$$\therefore I_g = -\frac{0.2}{910} = -0.0002198 \text{ A}.$$

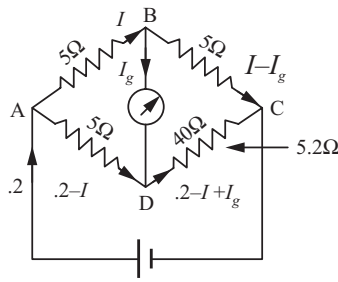


FIG. 6.10

ELECTRICAL MEASUREMENTS

6.7 CAREY FOSTER BRIDGE

Description. The Carey Foster bridge is a form of Wheatstone's bridge. It consists of a uniform wire AB of length 1 metre stretched on a wooden board (Fig. 6.11).

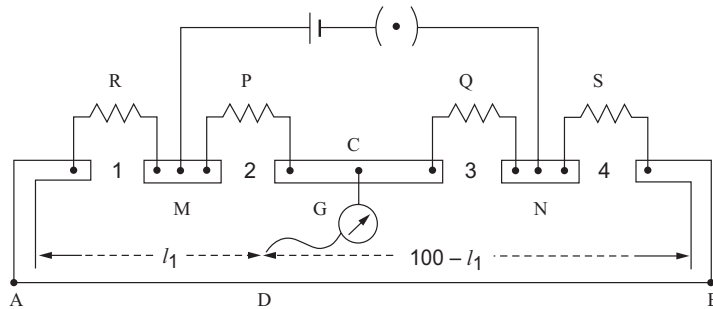


FIG. 6.11

Two equal resistances P and Q are connected in gaps 2 and 3. The unknown resistance R is connected in gap 1. A standard resistance S , of the same order of resistance as R , is connected in gap 4. A Leclanche cell is connected across MN . A galvanometer G is connected between the terminal C and a sliding contact maker D .

Theory. The contact maker is moved until the bridge is balanced. Let l_1 be the balancing length as measured from end A . Let α and β be the end resistances at A and B . Let ρ be the resistance per unit length of the wire.

From the principle of Wheatstone's bridge,

$$\frac{P}{Q} = \frac{R + \alpha + l_1\rho}{S + \beta + (100 - l_1)\rho} \quad \dots(1)$$

The resistances R and S are interchanged and the bridge is again balanced. The balancing length l_2 is determined from the same end A . Then,

$$\frac{P}{Q} = \frac{S + \alpha + l_2\rho}{R + \beta + (100 - l_2)\rho} \quad \dots(2)$$

Figs. 6.12 and 6.13 represent the equivalent Wheatstone's bridge circuit in the two cases.

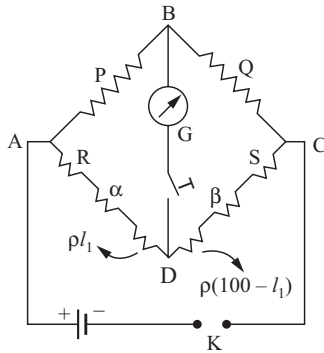


FIG. 6.12

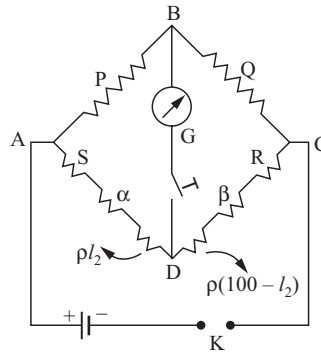


FIG. 6.13

From Eqns. (1) and (2),

$$\frac{R + \alpha + l_1 \rho}{S + \beta + (100 - l_1) \rho} = \frac{S + \alpha + l_2 \rho}{R + \beta + (100 - l_2) \rho} \quad \dots(3)$$

Adding 1 to both sides of Eq. (3),

$$\frac{R + \alpha + l_1 \rho + S + \beta + 100 \rho - l_1 \rho}{S + \beta + (100 - l_1) \rho} = \frac{S + \alpha + l_2 \rho + R + \beta + 100 \rho - l_2 \rho}{R + \beta + (100 - l_2) \rho}$$

$$\therefore \frac{R + S + \alpha + \beta + 100 \rho}{S + \beta + (100 - l_1) \rho} = \frac{R + S + \alpha + \beta + 100 \rho}{R + \beta + (100 - l_2) \rho}$$

Since the numerators are equal, the denominators must be equal.

$$\therefore S + \beta + 100 \rho - l_1 \rho = R + \beta + 100 \rho - l_2 \rho \quad \dots(4)$$

$$\text{or} \quad S - l_1 \rho = R - l_2 \rho$$

$$\therefore R = S + \rho (l_2 - l_1) \quad \dots(5)$$

To find ρ . A standard resistance of $0.1 \, \Omega$ is connected in gap 1. A thick copper strip is connected in gap 4 *i.e.*, $R = 0.1 \, \Omega$ and $S = 0$. The balancing length l_1' is determined. The standard resistance and the thick copper strip are interchanged. The balancing length l_2' is determined.

$$\text{From Eq. (5),} \quad 0.1 = S + \rho (l_2' - l_1')$$

$$\text{or} \quad \rho = \frac{0.1}{(l_2' - l_1')}$$

Thus by knowing S and ρ , the unknown resistance R is calculated.

Determination of Resistivity

The resistance R of the given wire is determined using Carey Foster's Bridge. The length of the wire L is measured. The mean radius (r) of the wire is found with a screw gauge. Then, the resistivity of the material of the wire is calculated using the formula,

$$s = \frac{\pi r^2 R}{L} \text{ ohm-metre.}$$

Determination of the temperature coefficient of resistance

Let R_0 and R_t be the resistances of a wire at temperatures 0°C and $t^\circ\text{C}$. Then,

$$R_t = R_0(1 + \alpha t)$$

or

$$\alpha = \frac{R_t - R_0}{R_0 t} = \frac{1}{R_0} \frac{dR}{dt}$$

where α is the temperature coefficient of resistance of the material.

The increase of resistance per unit original resistance per degree rise of temperature is called temperature coefficient of resistance.

The given wire is wound non-inductively in the form of a double spiral on a glass tube. It is immersed in a beaker containing ice at 0°C . The resistance of the wire is determined as above. The resistance of the wire is determined at 10° , 20° , 30° , ... 100°C . A graph is drawn with temperature along the X-axis and resistance along the Y-axis (Fig. 6.14). A straight line is obtained.

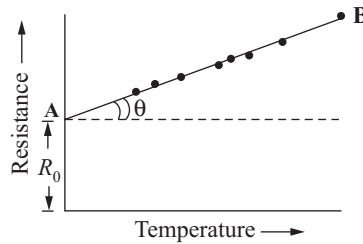


FIG. 6.14

$$\left. \begin{array}{l} \text{Slope of} \\ \text{the line} \end{array} \right\} = \tan \theta = \frac{dR}{dt}.$$

$$Y \text{ intercept} = R_0.$$

α is calculated using the formula, $\alpha = \frac{1}{R_0} \frac{dR}{dt}$

Note. Let R_1 and R_2 be the resistances at $t_1^\circ\text{C}$ and $t_2^\circ\text{C}$ respectively.

$$R_1 = R_0 [1 + \alpha t_1] \text{ and } R_2 = R_0 [1 + \alpha t_2]$$

$$\therefore \alpha = \frac{R_2 - R_1}{R_1 t_2 - R_2 t_1}$$

Example 1. In an experiment with Carey Foster bridge, the shift in the balance point is 5.4 cm when a thick copper strip and one ohm resistance are interchanged. The one ohm resistance is then replaced by an unknown resistance. Now the balance point shifts by 10 cm on interchanging. Calculate the unknown resistance.

Solution. $\rho = \frac{1}{5.4} \text{ ohm/cm}$

$$R = S + \rho(l_2 - l_1) = 0 + \frac{1}{5.4} \times 10 = 1.85 \text{ ohm.}$$

6.8 POTENTIOMETER

Principle. A potentiometer is a device for measuring or comparing potential differences. A potentiometer can be used to measure any electrical quantity which can be converted into a proportionate D.C potential difference.

It consists of a uniform wire AB of length 10 m stretched on a wooden board (Fig. 6.15). A steady current is passed through the wire AB with the help of a cell of EMF E . Let

ρ = resistance per unit length of potentiometer wire, and

I = steady current passing through the wire.

Let C be a variable point.

Let $AB = L$ and $AC = l$.

$$PD \text{ across } AB = L\rho I,$$

and $PD \text{ across } AC = l\rho I$

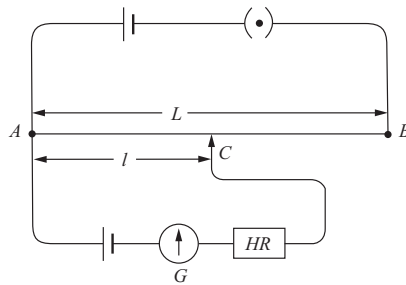


FIG. 6.15

$$\frac{PD \text{ across } AB}{PD \text{ across } AC} = \frac{L\rho I}{l\rho I} = \frac{L}{l}$$

$$\therefore PD \text{ across } AC = \frac{l}{L} \times PD \text{ across } AB$$

i.e., for a steady current passing through the potentiometer wire AB , the PD across any length is proportional to the length of the wire.

If a D.C. voltmeter is connected between A and the variable point C , it will be noted that the voltmeter registers greater values of PD's as the point C slides from A to B .

Calibration of Ammeter

Connect the ends of the potentiometer wire to the terminals of a storage cell through a key K_1 (Fig. 6.16). S is a standard cell. Connect the ammeter (A) to be calibrated in series with a battery, key K_2 , a rheostat and a standard resistance R . When a current I passes through the standard resistance R , the PD across R is IR . This potential drop is measured with the help of potentiometer.

Connect 1 and 3 and balance the EMF of the standard cell against the potentiometer. Find the balancing length (l) from A . The PD per cm of the potentiometer $= E/l$.

Connect 2 and 3. Adjust the rheostat so that the ammeter reads a value A_1 . Balance the PD across R against the potentiometer and find the balancing length l_1 .

$$PD \text{ across } R = El_1/l$$

$$\text{Current through } R = El_1/(IR).$$

$$\text{Correction to ammeter reading} = (El_1/IR) - A_1$$

Similarly, the corrections for other ammeter readings are determined. A calibration curve is plotted for ammeter, taking ammeter readings on X -axis and corrections on Y -axis (Fig. 6.17).

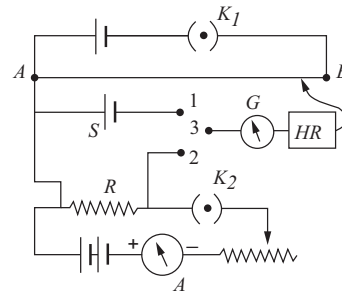


FIG. 6.16

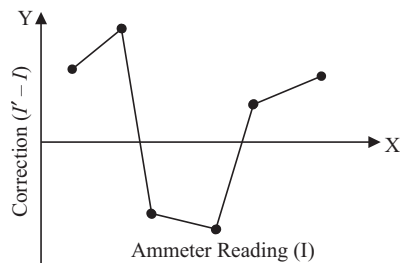


FIG. 6.17

Calibration of Voltmeter (Low range)

The connections are made as shown in Fig. 6.18. The voltmeter is connected parallel to R . Let l be the balancing length for the standard cell. The PD across R is balanced against the potentiometer. Let l_1 be the balancing length when the voltmeter reads V_1 .

$$PD \text{ across } R = El_1/l$$

$$\text{Correction to voltmeter} = (El_1/l) - V_1.$$

The experiment is repeated for various readings of the voltmeter and a calibration graph is drawn (Fig. 6.19).

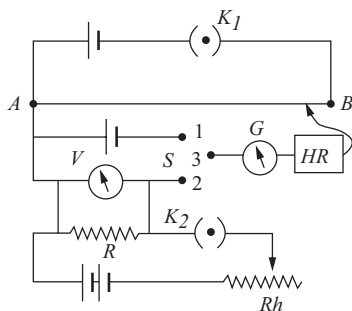


FIG. 6.18

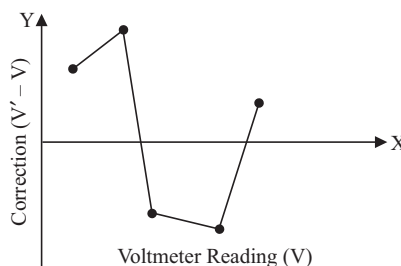


FIG. 6.19

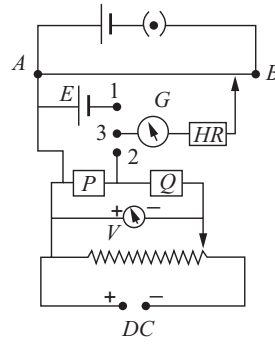
Calibration of voltmeter (High range)

Connections are made as shown in Fig. 6.20. Take suitable high resistances in P and Q such that the PD across P does not exceed the PD across the potentiometer. The balancing length l for the standard cell is determined first. Then the PD across P is balanced against the potentiometer and the balancing length l_1 is determined.

$$\text{PD across } P = El_1/l$$

$$\text{PD across } P + Q = \left(\frac{P+Q}{P} \right) \left(\frac{El_1}{l} \right)$$

$$\text{Correction to voltmeter} = \left(\frac{P+Q}{P} \right) \left(\frac{El_1}{l} \right) - V_1$$

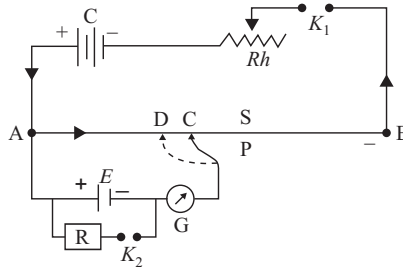
**FIG. 6.20**

The experiment is repeated for various readings of the voltmeter. A calibration curve is plotted for voltmeter, taking voltmeter readings on X -axis and corrections on Y -axis.

Determination of the Internal Resistance of a cell

Circuit Diagram. AB is the potentiometer wire (Fig. 6.21). A steady current is passed through the wire with the help of a battery. E is the cell whose internal resistance is to be measured. A resistance box R is connected across the cell through a key K_2 .

Procedure: Closing key K_1 , a balancing point is obtained on the potentiometer wire with K_2 open. The balancing length $l_1 (= AC)$ now is a measure of the EMF E of the cell.

**FIG. 6.21**

$$E \propto l_1 \quad \dots(1)$$

The key K_2 is closed. A resistance R is introduced in the box. Without disturbing rheostat Rh , the balancing length $l_2 (= AD)$ is measured. This is a measure of the PD V of the cell. Then

$$V \propto l_2 \quad \dots(2)$$

$$\text{From Eqns. (1) and (2),} \quad \frac{E}{V} = \frac{l_1}{l_2} \quad \dots(3)$$

Let E be the EMF of the cell and r the internal resistance. Let V be the PD across the cell when supplying a current I through the external resistance R .

$$\text{Then,} \quad V = IR \quad \dots(4)$$

$$\text{and} \quad E = I(R + r) \quad \dots(5)$$

$$\therefore \quad \frac{E}{V} = \frac{R+r}{R} = 1 + \frac{r}{R} \quad \dots(6)$$

$$\text{Comparing Eqns. (3) and (6),} \quad 1 + \frac{r}{R} = \frac{l_1}{l_2} \quad \dots(7)$$

$$\therefore \quad r = \left(\frac{l_1 - l_2}{l_2} \right) R \quad \dots(7)$$

Hence r is calculated.

Measurement of Resistance of a coil with a Potentiometer

Procedure: Connections are made as shown in Fig. 6.22.

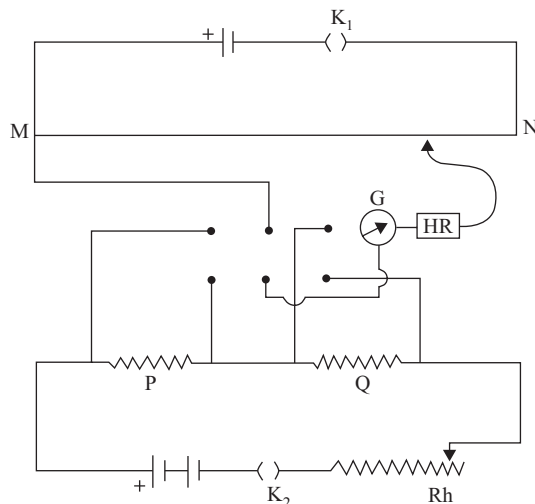


FIG. 6.22

The positive and negative poles of a storage battery are connected to the ends M and N of the potentiometer wire including a key K_1 in the circuit. The two resistances P and Q to be compared are joined in series with another battery, a rheostat and a plug key K_2 . By means of the double throw switch, the two resistances P and Q can be successively included in the galvanometer circuit. Care should be taken to see that the high potential terminals of the resistances P and Q are connected to the high potential terminal M of the potentiometer wire.

- (i) The P.D across the resistance P is first balanced on the potentiometer. The balancing length l_1 is found out.
- (ii) Keeping the current passing through P and Q unaltered, the P.D across Q is balanced on the potentiometer wire. The balancing length in this case l_2 is found out.

Theory: Let i be the current passing through the resistances P and Q . The potential differences across P and Q are iP and iQ respectively.

Now $iP \propto l_1$ and $iQ \propto l_2$.

By division,

$$\frac{P}{Q} = \frac{l_1}{l_2}$$

The experiment is repeated by altering the current passing through the resistances by means of the rheostat. The mean value of the ratio between the resistances is determined.

If one of the resistances is known, the value of the other resistance can be found out.

$$Q = P \left(\frac{l_2}{l_1} \right).$$

Determination of Resistivity

The length of the wire L is measured.

The mean radius (r) of the wire is found with a screw gauge.

The resistivity of the material of the wire is calculated using the formula,

$$s = Q \left(\frac{\pi r^2}{L} \right) \Omega m.$$

NETWORK ANALYSIS AND NETWORK THEOREMS

6.12 INTRODUCTION

Any electrical circuit containing resistances, capacitances, inductances and *generators*, (i.e., sources of emf) is known as electrical network. The *circuit elements* are of two types: (a) *passive* and (b) *active*. The inductor, resistor and capacitor with two terminals are regarded as *passive elements* as they are not fundamental sources of power. The *active elements* are those which feed the power into the given circuits. These are of two types: (a) *voltage source* and (b) *current source*.

An *ideal voltage source* is an active device which can maintain a constant voltage across its terminals irrespective of the current supplied by it. It is also called *constant voltage source*. It is always equal to the *open-circuit voltage*. [Fig. 6.27(a)].

A real voltage generator has a finite internal impedance. Therefore the potential difference across its terminals decreases with increase of load impedance [Fig. 6.27 (b)].

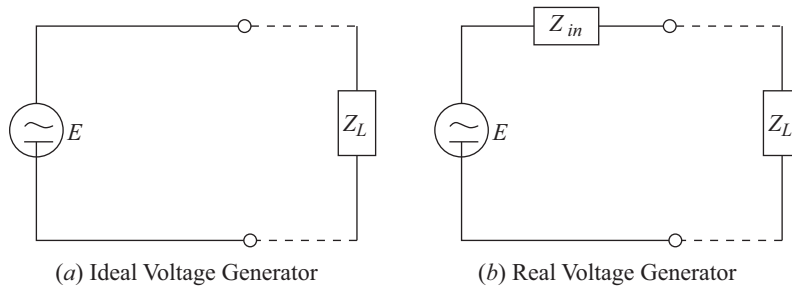


FIG. 6.27

A *constant current source* is an active device which can supply the constant current to any load resistance that is connected across its terminals [Fig. 6.28 (a)].

A real current generator is a power source which contains a finite internal parallel impedance [Fig. 6.28 (b)].

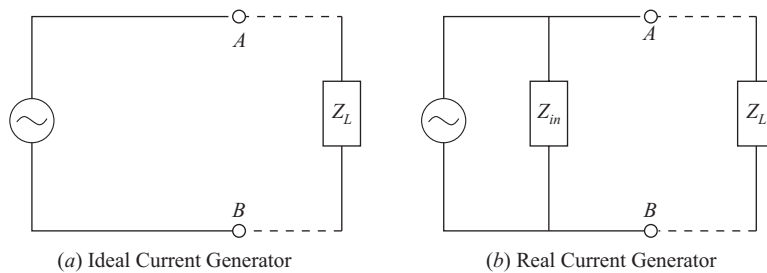


FIG. 6.28

Networks - Some Important Definitions

Fig. 6.29 represents a *network*.

Junction (or Node). The point where two or more branches meet is called a *junction* (or *node*), *a* and *b* are junctions.

Branch. In a network any group of series elements having two terminals is called a *branch*. The current in a branch remains same at each point.

An element joining two nodes, such as *a* to *b* in Fig. 6.29, is called a *branch*.

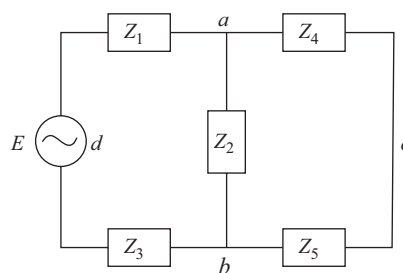


FIG. 6.29

A set of branches forming a closed path in a network is called a *mesh*.

Loop. Any closed path in a network is called a *loop*, *acba* and *abda* are loops.

Active and Passive Networks. The networks containing sources of e.m.f. are called active networks. Networks containing no sources of e.m.f. are called passive networks.

Linear and Non-Linear Networks

The network is said to be a linear network if it consists of linear impedances. A linear impedance is any impedance that obeys Ohm's law. In a *non-linear network*, the relation between voltage and current will be non-linear.

Network Theorems

Network problems can be solved by the application of Ohm's law and Kirchoff's laws. But in complicated circuits, calculations based on these laws are very lengthy and tedious. However, they can be easily solved by the application of certain theorems called *network theorems*.

6.13 SUPERPOSITION THEOREM

Statement. In any linear network containing impedances and more than one source of e.m.f. the current flowing in any element is equal to algebraic sum of the currents that would separately flow in that element if each source of e. m. f. were considered separately, all the other sources being replaced at that time by their internal impedances.

Proof. Let us consider the network with two generators of emf's E_1 and E_2 with internal impedances Z_1 and Z_2 respectively (Fig. 6.30). Let the currents due to E_1 and E_2 acting together be I_1 and I_2 [Fig. 6.30 (a)]. Let the currents due to e.m.f. E_1 acting alone be I_1' and I_2' [Fig. 6.30 (b)]. Let the currents due to E_2 acting alone be I_1'' and I_2'' [Fig. 6.30 (c)].

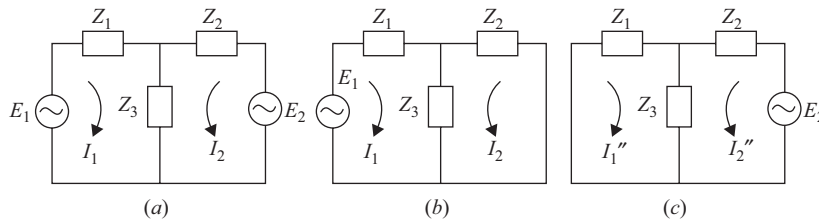


FIG. 6.30

Applying Kirchoff's II Law to Fig. 6.30, the mesh equations are

$$E_1 = (Z_1 + Z_3) I_1 + Z_3 I_2 \quad \dots(1)$$

$$E_2 = Z_3 I_1 + (Z_2 + Z_3) I_2 \quad \dots(2)$$

When E_1 is considered to act alone [Fig. 6.30 (b)], the mesh equations are

$$E_1 = (Z_1 + Z_3) I_1' + Z_3 I_2' \quad \dots(3)$$

$$0 = Z_3 I_1' + (Z_2 + Z_3) I_2' \quad \dots(4)$$

When E_2 is considered to act alone [Fig. 6.30 (c)], the mesh equations are

$$0 = (Z_1 + Z_3) I_1'' + Z_3 I_2'' \quad \dots(5)$$

$$E_2 = Z_3 I_1'' + (Z_2 + Z_3) I_2'' \quad \dots(6)$$

Adding Eqs. (3) and (5),

$$E_1 = (Z_1 + Z_3) (I_1' + I_1'') + Z_3 (I_2' + I_2'') \quad \dots(7)$$

Adding Eqs. (4) and (6),

$$E_2 = (Z_2 + Z_3)(I_2' + I_2'') + Z_3(I_1' + I_1'') \quad \dots(8)$$

Eqs. (7) and (8) will be identical with Eqs. (1) and (2) respectively, if

$$I_1 = I_1' + I_1'' \quad \text{and} \quad I_2 = I_2' + I_2''$$

Thus the theorem is proved.

Example 1. Find the current through $3\ \Omega$ resistor of the following network (Fig. 6.31) using superposition theorem.

Solution. Replace the battery of e.m.f. 10 V by its internal resistance $1\ \Omega$ and obtain the circuit in Fig. 6.32.

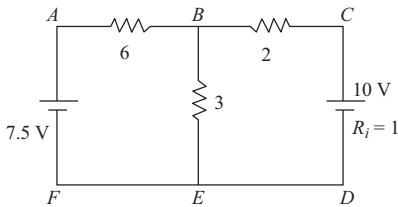


FIG. 6.31

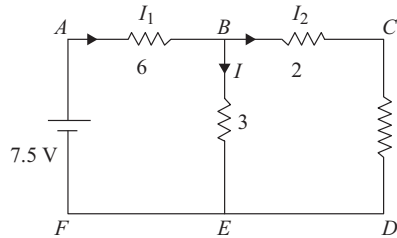


FIG. 6.32

Let I_1' , I_2' and I' be the currents through $6\ \Omega$, $2\ \Omega$ and $3\ \Omega$ respectively.

Total resistance offered to the voltage source of e.m.f. 7.5 V

$$= 6 + \frac{3 \times (2 + 1)}{3 + (2 + 1)} = 7.5\ \Omega$$

Current drawn from the cell is

$$I_1' = \frac{7.5}{7.5} = 1\text{ A}$$

Current in BE is

$$\begin{aligned} I' &= \frac{I_1' \times (2 + 1)}{3 + (2 + 1)} \\ &= \frac{1 \times 3}{6} = 0.5\text{ A} \end{aligned}$$

Now replace the battery of e.m.f. 7.5 V by its resistance. Here the internal resistance is zero. Therefore, short the battery of e.m.f. 7.5 V and obtain the circuit in Fig. 6.33.

Let I_1'' , I_2'' and I'' be currents through $6\ \Omega$, $2\ \Omega$ and $3\ \Omega$ respectively.

Total resistance offered to the voltage source of e.m.f. 10 V

$$= \frac{6 \times 3}{6 + 3} + 2 + 1 = 2 + 2 + 1 = 5\ \Omega$$

Current drawn from the battery is

$$I_2'' = \frac{10}{5} = 2\text{ A}$$

$\therefore$ Current through $3\ \Omega$ resistor I'' in this circuit

$$\begin{aligned} &= \frac{I_2'' \times 6}{6 + 3} = \frac{2 \times 6}{9} \\ &= \frac{12}{9} = 1.33\text{ A} \end{aligned}$$

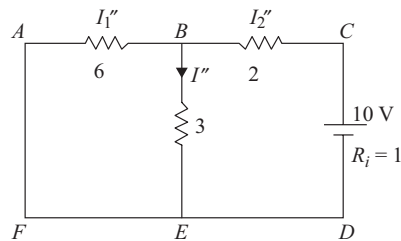


FIG. 6.33

According to superposition theorem, the total current through $3\ \Omega$ resistor is

$$I = I' + I'' = 0.5\text{ A} + 1.33\text{ A} = 1.83\text{ A}$$

6.14 THEVENIN'S THEOREM

Statement. The current in a load impedance connected between two terminals of a network of generators and linear impedances is same as due to a single voltage generator of e.m.f. equal to open circuit voltage between the same terminals of the network and internal impedance equal to the impedance of the network between the same terminals of the network when all the generators in the network have been replaced by their internal impedances.

The schematic representation of Thevenin's theorem is shown in Fig. 6.34.

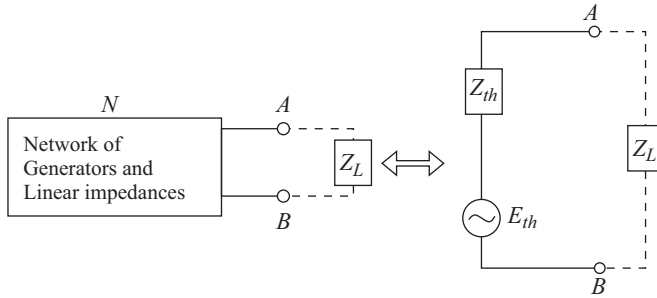


FIG. 6.34

N is a network containing a number of generators and linear impedances with output terminals A and B . Let E_{th} be the open-circuit voltage across A and B . Let Z_{th} be the impedance measured between the terminals A and B when all the generators have been replaced by their respective internal impedances. The network N will produce the same current in an external load impedance Z_L connected across A and B as a single voltage generator of e.m.f. E_{th} and internal impedance Z_{th} would do.

Proof. Consider a network containing resistive impedances Z_1 , Z_2 and Z_3 and one generator of e.m.f. E and internal impedance zero (i.e., ideal generator) [Fig. 6.35 (a)]. I_1 is current supplied by the source. I_2 is the current flowing through the load impedance Z_L .

Applying Kirchoff's II law to meshes I and II, we get

$$I_1 Z_1 + (I_1 - I_2) Z_3 = E$$

$$\text{or } I_1 (Z_1 + Z_3) - I_2 Z_3 = E \quad \dots(1)$$

$$\text{and } I_2 Z_2 + I_2 Z_L - (I_1 - I_2) Z_3 = 0$$

$$I_2 (Z_2 + Z_L + Z_3) = I_1 Z_3 \quad \dots(2)$$

$$\text{or } I_1 = \frac{I_2 (Z_2 + Z_3 + Z_L)}{Z_3}$$

Substituting this value of I_1 in Eqn. (1), we get

$$I_2 \frac{(Z_2 + Z_3 + Z_L)}{Z_3} (Z_1 + Z_3) - I_2 Z_3 = E$$

$$\text{or } I_2 [(Z_2 + Z_3 + Z_L) (Z_1 + Z_3) - Z_3^2] = E Z_3$$

$$\text{or } I_2 = \frac{E Z_3}{Z_2 (Z_1 + Z_3) + Z_1 Z_3 + Z_L (Z_1 + Z_3)}$$

$$\therefore I_2 = \frac{E Z_3 / (Z_1 + Z_3)}{Z_2 + \left(\frac{Z_1 Z_3}{Z_1 + Z_3} \right) + Z_L} \quad \dots(3)$$

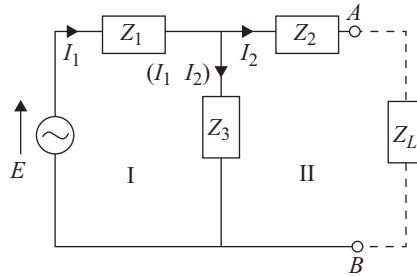


FIG. 6.35 (a)

Again, from Fig. 6.35 (b), the open-circuit voltage at terminals A, B (i.e., when Z_L is removed)

$$E_{th} = \frac{EZ_3}{Z_1 + Z_3} \quad \dots(4)$$

Further if E is replaced by its zero internal impedance [Fig. 6.35 (c)], the impedance of network between A and B is

$$\begin{aligned} Z_{th} &= Z_2 + \frac{1}{\frac{1}{Z_1} + \frac{1}{Z_3}} \\ &= Z_2 + \frac{Z_1 Z_3}{Z_1 + Z_3} \end{aligned} \quad \dots(5)$$

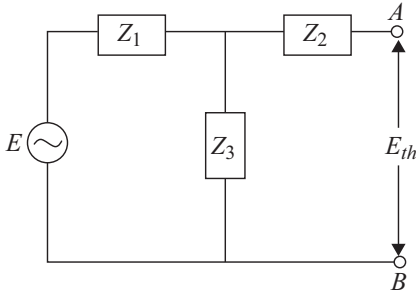


FIG. 6.35 (b)

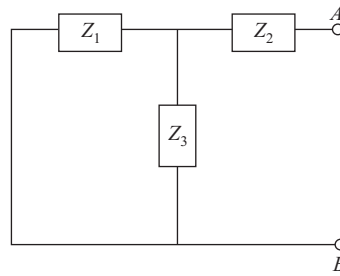


FIG. 6.35 (c)

Thus Eq. (3) is equivalently written as

$$I_2 = \frac{E_{th}}{Z_{th} + Z_L} \quad \dots(6)$$

This proves Thevenin's theorem. The equivalent circuit is shown in Fig. 6.35 (d).

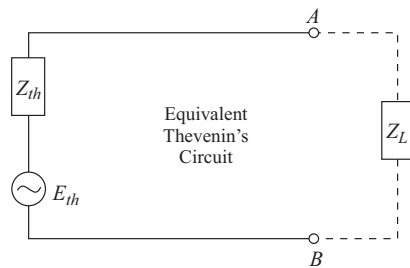


FIG. 6.35 (d)

Example 1. Use Thevenin's theorem to find the current in load resistance of $100\ \Omega$ in the given circuit [Fig. 6.36],

Solution. (i) First of all we disconnect load resistance of $100\ \Omega$ [Fig. 6.37 (a)].

(ii) **To determine Thevenin's e.m.f. E_{th} .** If I is the current in $10\ \Omega$ resistor, then current in $5\ \Omega$ resistor is also I since terminals A and B are open circuited.

According to Kirchhoff's II law, the equation for closed loop is

$$10I + 5I = 10$$

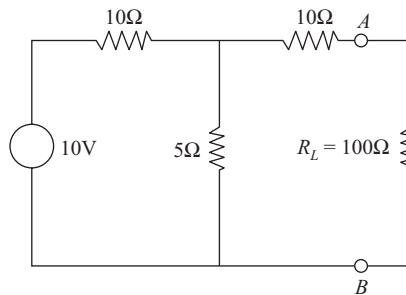


FIG. 6.36

$$\therefore I = \frac{10}{15} = \frac{2}{3} \text{ A}$$

Voltage across AB = Voltage across 5Ω resistor = $(2/3) \times 5 = (10/3) \text{ V}$

(iii) **To determine Thevenin's Resistance R_{th} .** Now the source of e.m.f. is replaced by its internal resistance which is zero in this case [Fig. 6.37 (b)].

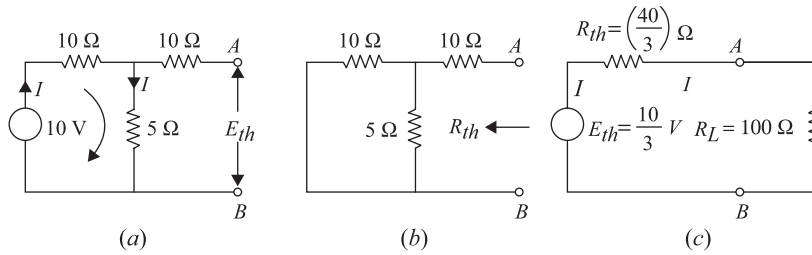


FIG. 6.37

As viewed across A, B , 10Ω and 5Ω resistances are in parallel and then 10Ω resistor is in series.

$$R_{th} = 10 + \frac{10 \times 5}{10 + 5} = 10 + \frac{50}{15} = 10 + \frac{10}{3} = \frac{40}{3} \Omega$$

The Thevenin's circuit is shown in Fig. 6.37 (c).

$\therefore$ The current through load R_L is

$$I_L = \frac{\text{Voltage}}{\text{Resistance}} = \frac{E_{th}}{R_{th} + R_L} = \frac{\left(\frac{10}{3}\right)}{\left(\frac{40}{3} + 100\right)} = \frac{1}{34} \text{ A}$$

6.15 NORTON'S THEOREM

Statement. Any two terminal active network, consisting of linear impedances and generators may be replaced by a current generator in parallel with an impedance. The current is equal to that obtained through a short circuit across the two terminals and the impedance is that viewed across the two terminals when all the generators are replaced by their internal impedances.

The schematic representation of Norton's theorem is shown in Fig. 6.38.

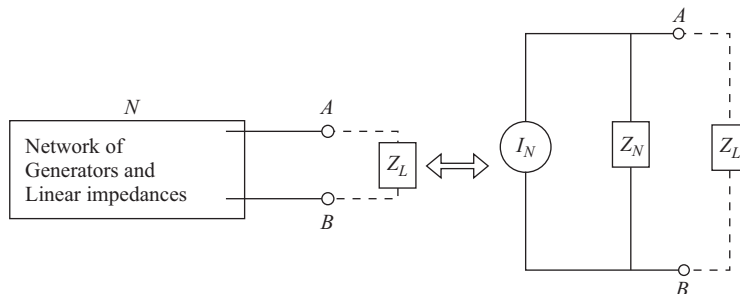
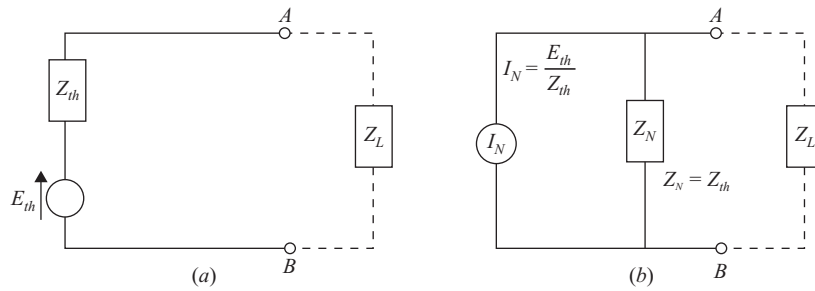


FIG. 6.38

This theorem is useful in reducing a complicated network into a *simple parallel circuit* consisting of an *ideal current source* and a parallel impedance. Hence the theorem finds its utility where current sources are of major interest than e.m.f.'s. e.g., in transistor circuits.

Proof. Consider the Thevenin's and Norton's equivalent circuits of same network shown in Fig. 6.39 (a) and (b).


FIG. 6.39

The current I_2 for Thevenin's equivalent circuit in load Z_L is

$$I_2 = \frac{E_{th}}{Z_{th} + Z_L} \quad \dots(1)$$

The current I_L for Norton's equivalent circuit in load Z_L is

$$I_L = \frac{I_N Z_N}{Z_N + Z_L}$$

But $Z_N = Z_{th}$ and $I_N = \frac{E_{th}}{Z_{th}}$

$$I_L = \frac{E_{th}}{Z_{th} + Z_L}$$

$$\therefore I_2 = I_L \quad \dots(2)$$

The current across the load impedance Z_L given by two equivalent circuits are equal.

Thus Norton's theorem is proved.

Example 1. Draw Thevenin's and Norton's equivalent circuits for following network of resistances (Fig. 6.40). Calculate the current in the load in each case.

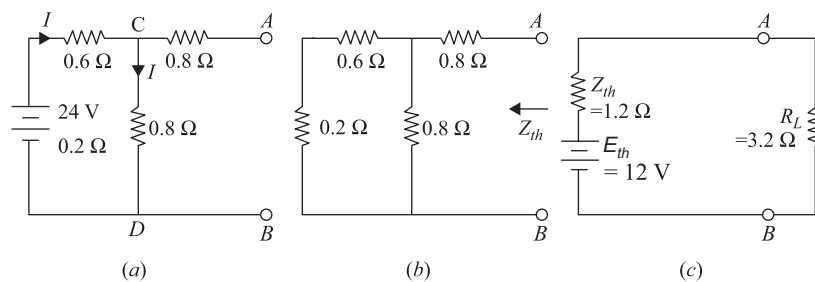
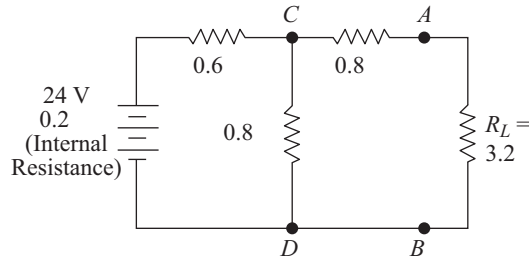
Solution. (i) We first remove R_L and leave A and B in open circuit.

Applying Kirchoff's second law to mesh of circuit [Fig. 6.41 (a)],

$$0.6 I + 0.8 I + 0.2 I = 24 \text{ or } I = 15 \text{ A}$$

$$E_{th} = \text{voltage across } CD = 15 \times 0.8 = 12 \text{ V}$$

(ii) Thevenin's internal impedance Z_{th} . The battery is removed and its internal resistance 0.2Ω is left behind [Fig. 6.41 (b)]. The internal impedance as viewed from A and B is the Thevenin's impedance given by


FIG. 6.41

FIG. 6.40

$$Z_{th} = 0.8 + \frac{(0.8)(0.6 + 0.2)}{0.8 + (0.6 + 0.2)} = 0.8 + 0.4 = 1.2 \Omega$$

The Thevenin's equivalent circuit is shown in Fig. 6.41 (c).

(iii) The current in load R_L when connected between AB is

$$I_L = \frac{12 \text{ Volt}}{1.2 \Omega + 3.2 \Omega} = \frac{12 \text{ Volt}}{4.4 \Omega} = \frac{30}{11} \text{ A}$$

Norton's Equivalent Circuit. The Norton's equivalent circuit may be found from Thevenin's equivalent circuit. The short-circuit current is found by shorting the terminal A and B together.

$$I_N = \frac{E_{th}}{Z_{th}} = \frac{12}{1.2} = 10 \text{ A}$$

The Norton's equivalent circuit is shown in Fig. 6.41 (d).

When the load $R_L = 3.2 \Omega$ is connected across AB , the current in the load is

$$I_L = \frac{Z_N}{Z_N + R_L} I_N = \frac{1.2}{1.2 + 3.2} \times 10 = \frac{12}{4.4} = \frac{30}{11} \text{ A}$$

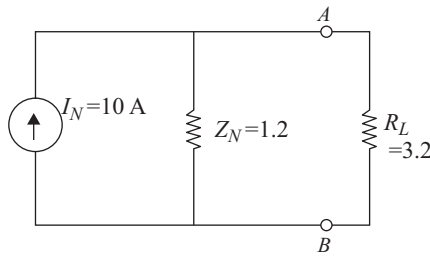


FIG. 6.41 (d)

6.16 MAXIMUM POWER TRANSFER THEOREM

Statement. The maximum power will be delivered to a load by a generator when

- resistive components of the impedances of load and generator are equal, and
- the reactive components of the impedances of generator and load are equal in magnitude but opposite in sign.

Proof. Consider a voltage generator of e.m.f. E' and complex internal impedance Z' and a load of complex impedance Z_L , connected across the generator (Fig. 6.42).

Now, we have

$$Z' = R' + jX' \quad \dots(1)$$

and

$$Z_L = R_L + jX_L \quad \dots(2)$$

Here, R' and R_L are the resistive components and X' and X_L are the reactive components of Z' and Z_L respectively.

The current I flowing in the circuit is

$$I = \frac{E'}{Z' + Z_L} = \frac{E'}{(R' + R_L) + j(X' + X_L)} \quad \dots(3)$$

The power delivered to the load is,

$$P = I^2 R_L = \frac{(E')^2 R_L}{(R' + R_L)^2 + (X' + X_L)^2} \quad \dots(4)$$

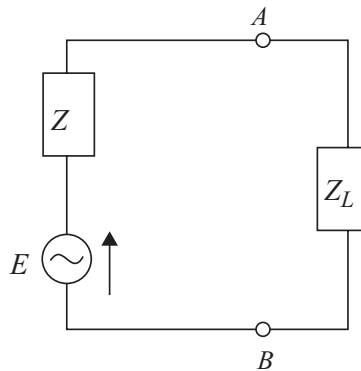


FIG. 6.42

Now let us consider the variation of P with X_L . The power P will be maximum when $(\partial P / \partial X_L) = 0$

$$\frac{\partial P}{\partial X_L} = - \frac{2 (E')^2 R_L (X_L + X')}{[(R_L + R')^2 + (X_L + X')^2]^2}$$

or $X_L = -X'$... (5)

Thus if X' is inductive then X_L must be capacitive and of the same magnitude as X' .

When this condition is satisfied, the maximum power is

$$P_{max} = \frac{(E')^2 R_L}{(R_L + R')^2} \quad \dots (6)$$

Again, consider the variation of P_{max} with R_L . The power P_{max} will be maximum when,

$$\frac{\partial P_{max}}{\partial R_L} = \frac{(E')^2 (R_L + R')^2 - 2 (E')^2 R_L (R_L + R')}{(R_L + R')^4} = 0$$

or $R_L = R'$... (7)

Eq. (7) shows that maximum power will be delivered to the load when the resistive components of generator and load are equal.

The expression for the maximum power may then be written from Eq. (6), as

$$P_{max} = \frac{E'^2 R_L}{(2R_L)^2} = \frac{(E')^2}{4R_L} \quad \dots (8)$$

Thevenin's Theorem (for dc Circuits)

Statement. The current in a load resistance connected between two terminals of a network of generators and linear resistances is same as due to a single voltage generator of e.m.f. equal to open circuit voltage between the same terminals of the network and internal resistance equal to the resistance of the network between the same terminals of the network when all the generators in the network have been replaced by their internal resistances (Fig. 6.43).

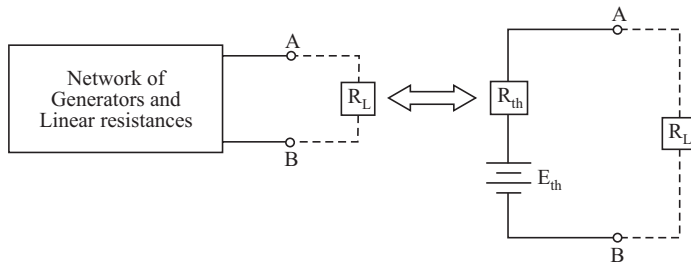


FIG. 6.43

Proof. Consider a network containing resistances R_1 , R_2 and R_3 and one generator of e.m.f. E and internal resistance zero (Fig. 6.44).

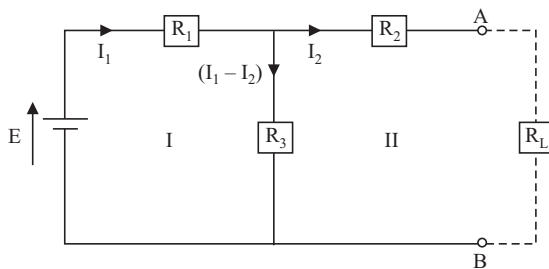


FIG. 6.44

Applying KVL to meshes I and II, we get

$$\begin{aligned} I_1 R_1 + (I_1 - I_2) R_3 &= E \\ \text{i.e., } I_1 (R_1 + R_3) - I_2 R_3 &= E \end{aligned} \quad \dots(1)$$

$$\begin{aligned} \text{and } I_2 R_2 + I_2 R_L - (I_1 - I_2) R_3 &= 0 \\ \text{i.e., } I_2 (R_2 + R_L + R_3) - I_1 R_3 &= 0 \end{aligned} \quad \dots(2)$$

From Eq. (2),
$$I_1 = \frac{I_2 (R_2 + R_3 + R_L)}{R_3}$$

Substituting this value of I_1 in Eq. (1), we get

$$I_2 [(R_2 + R_3 + R_L) (R_1 + R_3) - R_3^2] = E R_3$$

This gives
$$I_2 = \frac{E R_3}{R_2 (R_1 + R_3) + R_1 R_3 + R_L (R_1 + R_3)} \quad \dots(3)$$

This is the current flowing through load resistance R_L .

Equation (3) may be put in the following form:

$$I_2 = \frac{E R_3 / (R_1 + R_3)}{R_2 + \left(\frac{R_1 R_3}{R_1 + R_3} \right) + R_L} \quad \dots(4)$$

This current is same as would flow in R_L if R_L were connected to a generator of e.m.f. $\frac{E R_3}{R_1 + R_3}$ and internal resistance $R_2 + \left(\frac{R_1 R_3}{R_1 + R_3} \right)$.

Now let us solve the problem by applying Thevenin's theorem.

(i) To find Thevenin voltage E_{th} :

R_L is removed (i.e., terminals A and B are open circuited) [Fig. 6.45].

The current in R_1 and R_3 is

$$I_1 = \frac{E}{R_1 + R_3} \quad \dots(5)$$

i.e., voltage across R_3 is

$$V = I_1 R_3 = \frac{E R_3}{R_1 + R_3}$$

As terminals A and B are open circuited, there is no current in R_2 and the p.d. E_{th} across A and B is

$$E_{th} = V = \frac{E R_3}{R_1 + R_3} \quad \dots(6)$$

(ii) To find the Thevenin's resistance R_{th} .

E is replaced by its zero internal resistance (Fig. 6.46).

The resistance of network between A and B is

$$R_{th} = R_2 + \frac{1}{\frac{1}{R_1} + \frac{1}{R_3}} = R_2 + \frac{R_1 R_3}{R_1 + R_3} \quad \dots(7)$$

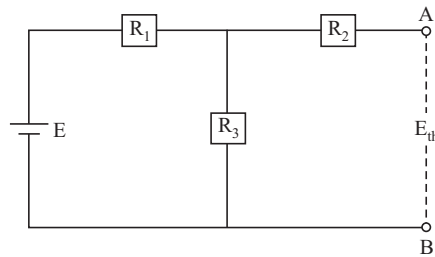


FIG. 6.45

Thus Eq. (4) is equivalently written as

$$I_2 = \frac{E_{th}}{R_{th} + R_L}$$

This proves Thevenin's theorem.

Fig. 6.47 shows the equivalent circuit.

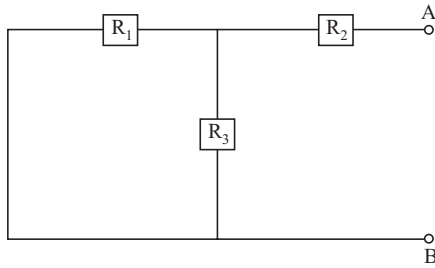


FIG. 6.46

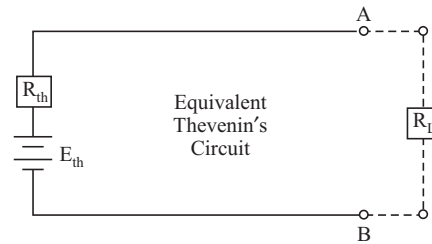


FIG. 6.47

6.17 MAXWELL'S LOOP CURRENT METHOD

This method employs a system of *loop* or *mesh* currents instead of *branch* currents (as in Kirchhoff's laws). Here, the currents in different meshes are assigned continuous paths so that they do not split at a junction into branch currents. If a network has a large number of voltage sources, it is useful to use mesh analysis. This method consists of writing loop voltage equations by Kirchhoff's voltage law in terms of unknown loop currents.

Consider a three-mesh circuit depicted in Fig. 6.48.

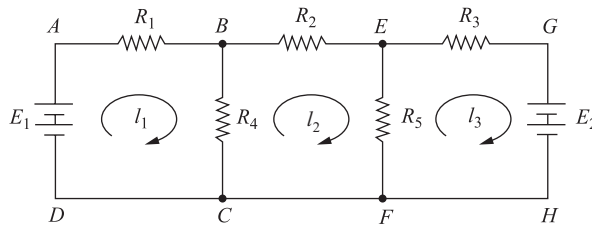


FIG. 6.48

The network has two batteries (voltage sources) and five resistors. There are three meshes traversed by mesh currents I_1 , I_2 and I_3 respectively. Current through R_4 (when considered as a part of the first loop) is $(I_1 - I_2)$ and that through R_5 is $(I_2 - I_3)$. However, when R_4 is considered part of the second loop, current through it is $(I_2 - I_1)$. Similarly, when R_5 is considered part of the third loop, current through it is $(I_3 - I_2)$. By applying KVL to the three loops, the equations may be written as follows:

Loop 1. $E_1 - I_1 R_1 - R_4 (I_1 - I_2) = 0$

or $I_1 (R_1 - R_4) - I_2 R_4 - E_1 = 0$

Loop 2. $-I_2 R_2 - R_5 (I_2 - I_3) - R_4 (I_2 - I_1) = 0$

or $I_1 R_4 - I_2 (R_2 + R_4 + R_5) + I_3 R_5 = 0$

Loop 3. $-I_3 R_3 - E_2 - R_5 (I_3 - I_2) = 0$

or $I_2 R_5 - I_3 (R_3 + R_5) - E_2 = 0$

The above three equations can be solved not only to find loop currents but branch currents as well.

Example 1. In a network given below, calculate the mesh currents I_1 , I_2 and I_3 by the method of determinants. Hence obtain the voltages V_1 and V_2 and also the voltage across the $6\ \Omega$ resistor. (Fig. 6.49).

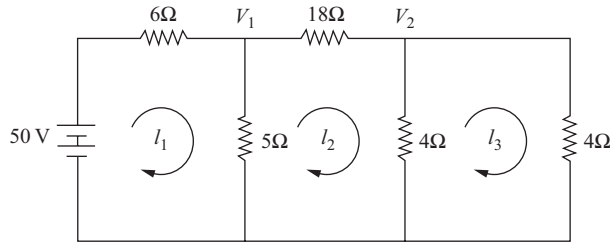


FIG. 6.49

Solution. The three equations for the three meshes may be written as

$$50 = (6 + 5) I_1 - 5 I_2 \quad \dots(1)$$

$$0 = -5I_1 + (5 + 18 + 4) I_2 - 4I_3 \quad \dots(2)$$

$$0 = -4 I_2 + (4 + 4) I_3 \quad \dots(3)$$

These equations may be written in the form of determinants as

$$\begin{vmatrix} 50 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 11 & -5 & 0 \\ -5 & 27 & -4 \\ 0 & -4 & 8 \end{vmatrix} \begin{vmatrix} I_1 \\ I_2 \\ I_3 \end{vmatrix}$$

The determinant of resistance may be written as

$$\begin{aligned} \Delta_R = [R] &= 11 \begin{vmatrix} 27 & -4 \\ -4 & 8 \end{vmatrix} + 5 \begin{vmatrix} -5 & -4 \\ 0 & 8 \end{vmatrix} \\ &= 11(27 \times 8 - 4 \times 4) + 5(-40) = 2000 \end{aligned}$$

$$\Delta_1 = \begin{vmatrix} 50 & -5 & 0 \\ 0 & 27 & -4 \\ 0 & -4 & 8 \end{vmatrix} = 50 \begin{vmatrix} 27 & -4 \\ -4 & 8 \end{vmatrix} = 10000$$

$$\Delta_2 = \begin{vmatrix} 11 & 50 & 0 \\ -5 & 0 & -4 \\ 0 & 0 & 8 \end{vmatrix} = -50 \begin{vmatrix} -5 & -4 \\ 0 & 8 \end{vmatrix} = 2000$$

$$\Delta_3 = \begin{vmatrix} 11 & -5 & 50 \\ -5 & 27 & 0 \\ 0 & -4 & 0 \end{vmatrix} = 50 \begin{vmatrix} -5 & 27 \\ 0 & -4 \end{vmatrix} = 1000$$

Thus the currents are given by

$$I_1 = \frac{\Delta_1}{\Delta_R} = \frac{10000}{2000} = 5 \text{ A}$$

$$I_2 = \frac{\Delta_2}{\Delta_R} = \frac{2000}{2000} = 1 \text{ A}$$

$$I_3 = \frac{\Delta_3}{\Delta_R} = \frac{1000}{2000} = 0.5 \text{ A}$$

and

The voltage $V_1 = (I_1 - I_2) 5 = 4 \times 5 = 20 \text{ V}.$

$$V_2 = (I_2 - I_3) 4 = 0.5 \times 4 = 2 \text{ V}.$$

Voltage across $6\Omega = 6I_1 = 6 \times 5 = 30 \text{ V}.$

Application of Maxwell's cyclic current method to Wheatstone's bridge

According to this method, a cyclic current of some specific value is imagined to flow in each mesh and all the currents in various meshes are assumed to flow in the same direction i.e., clockwise or anticlockwise. The actual current flowing in any branch of the mesh is the algebraic sum of the cyclic currents flowing in the meshes which the branch separates. Further, the e.m.f. is taken as positive if it sends the current in the same direction in which the cyclic current is assumed to flow and negative if it sends the current in the opposite direction.

Let x , y and z be the currents flowing in the clockwise direction through the meshes (1) $ABDA$, (2) $BCDB$, and (3) $ADCEA$ (Fig. 6.50).

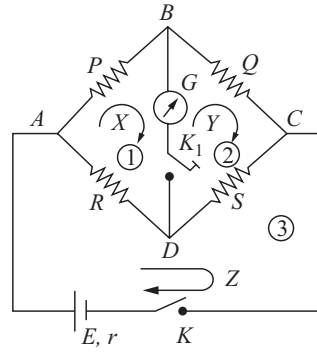


FIG. 6.50

Applying Kirchhoff's second law to the meshes (1), (2) and (3),

$$xP + (x - y)G + (x - z)R = 0 \quad \dots(1)$$

$$yQ + (y - z)S + (y - x)G = 0 \quad \dots(2)$$

and $(z - x)R + (z - y)S + zr = E \quad \dots(3)$

Collecting the coefficient of x , y and z in Eqs. (1) and (2),

$$x(P + G + R) - yG - zR = 0 \quad \dots(4)$$

and $-xG + y(Q + S + G) - zS = 0 \quad \dots(5)$

Multiplying Eq. (4) by S and Eq. (5) by R and subtracting, we get,

$$x[S(P + G + R) + RG] - y[GS + R(Q + S + G)] = 0$$

The current passing through the galvanometer $= (x - y)$. When the bridge is balanced, the current through the galvanometer is zero,

i.e., $x - y = 0$ or $x = y$

$\therefore SP + SG + SR + RG = GS + RQ + RS + RG$

or $SP = RQ$ or $\frac{P}{Q} = \frac{R}{S} \quad \dots(6)$

Eq. (6) is the necessary condition for a balanced Wheatstone bridge.

6.18 NODAL ANALYSIS WITH CURRENT SOURCES

A *node* is defined as a junction of two or more branches. If we now define one node of any network as a reference (that is, a point of zero potential or ground), the remaining nodes of the network will all have a fixed potential relative to this reference. For a network of N nodes, therefore, there will exist $(N - 1)$ nodes with a fixed potential relative to the assigned reference node. Equations relating these nodal voltages can be written by applying Kirchhoff's current law at each of the $(N - 1)$ nodes. To obtain the complete solution of a network, the resulting equations are solved for nodal voltages.

According to problem, the weight of lower wire is supported by upward magnetic repulsion.

$$\therefore \frac{\mu_0 i_a i_b}{2\pi d} = 0.073$$

$$\text{or } d = \frac{\mu_0 i_a i_b}{2\pi \times 0.073} = \frac{4\pi \times 10^{-7} \times 100 \times 20}{2\pi \times 0.073} = 5.48 \times 10^{-3} \text{ m.}$$

10.10 TORQUE ON A CURRENT LOOP IN A UNIFORM MAGNETIC FIELD

Consider a rectangular loop, height l and width b , carrying a current i (Fig. 10.24). The axis of the loop is perpendicular to the uniform magnetic field $\vec{B}$. The normal to the plane of the loop makes an angle θ with the direction of magnetic induction $\vec{B}$.

The forces acting on bottom and top sides 1 and 3 are $\vec{F}_1$ and $\vec{F}_3$. They are equal, opposite and act along the same line.

$$\begin{aligned} |\vec{F}_1| &= -|\vec{F}_3| = |i \vec{b} \times \vec{B}| \\ &= i b B \sin(90 - \theta) = i b B \cos \theta \quad \dots (1) \end{aligned}$$

By the vector product rule, this force is upward on the top side (3) and downward on the bottom side (1). The net force on the rigid loop from these elements is thus zero, as is the torque.

Forces acting on the vertical sides 2 and 4 are $\vec{F}_2$ and $\vec{F}_4$.

They are equal and opposite but do not act along the same line.

The common magnitude of $\vec{F}_2$ and $\vec{F}_4$ is $i l B$.

These two forces produce a torque

$$\tau = i l B b \sin \theta = i A B \sin \theta$$

where $A = lb = \text{area of the loop}$.

If there are N turns on the loop,

$$\tau = N i A B \sin \theta. \quad \dots (2)$$

We define the magnetic dipole moment of a plane current loop as

$$|\vec{\mu}| = \mu = \text{Area} \times \text{Current} = N i A$$

The direction of $\vec{\mu}$ is given by the right handed screw rule.

$$\therefore \vec{\tau} = \vec{\mu} \times \vec{B} = \mu B \sin \theta \quad \dots (3)$$

The *moving coil galvanometer* is based on this concept.

The D'Arsonval Moving Coil Galvanometer

Principle. It is an instrument used for the measurement of current. Its action is based on the torque acting on a current-carrying coil placed in a magnetic field.

- It is an aperiodic or dead beat galvanometer.

Construction. It consists of a rectangular coil (10 to 20 turns) of insulated fine Cu-wire wound on a light brass or aluminium frame (Fig. 10.25). The coil is suspended between the cylindrical pole pieces of a permanent horse-shoe magnet NS by means of a thin phosphor-bronze strip. The upper end of the strip is attached to a

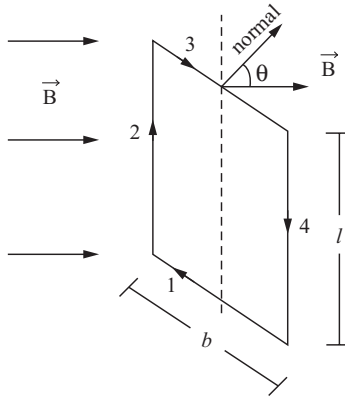


FIG. 10.24

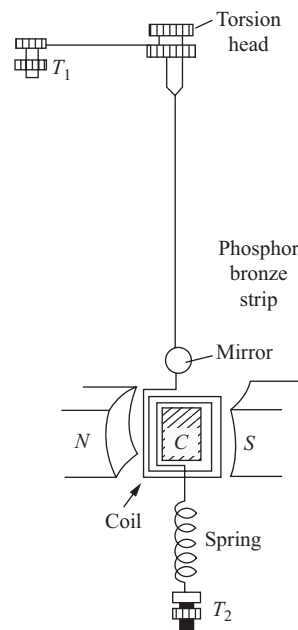


FIG. 10.25

torsion head. The lower end of the coil is attached to a spring of phosphor-bronze. A soft iron cylindrical core C is placed symmetrically within the coil without touching it. It concentrates the lines of force and thus makes the magnetic field between the pole-pieces *strong*. A small mirror is attached to the lower portion of the suspension strip. The mirror is used to reflect a beam of light on to a (translucent) scale (Fig. 10.26).

The current to be measured enters at one terminal T_1 and passes through the suspension, coil and spring and finally leaves at the second terminal T_2 .

Theory. Let B be the magnitude of the magnetic field induction due to the permanent magnet, A the area of the coil, and N the number of turns in it. The magnitude of the torque acting on the entire coil when carrying a current i is

$$\tau = NiBA \sin \theta$$

Here, θ is the angle which the normal to the plane of the coil makes with the direction of $\mathbf{B}$. The magnetic field $\mathbf{B}$ is made 'radial' by making the pole-pieces *cylindrical* (Fig. 10.27). In this case the normal to the plane of the coil is always at right angles to the magnetic field $\mathbf{B}$ so that $\theta = 90^\circ$. Hence the torque acting on the coil is

$$\tau = NiBA \sin 90^\circ = NiBA.$$

Let ϕ be the steady angular deflection of the coil. Then, the elastic torque developed in the suspension is $c\phi$, where c is the 'torsional constant' of the suspension. Therefore, we have

$$\tau = NiBA = c\phi$$

or

$$i = \frac{c}{NBA} \phi$$

or

$$i = k\phi$$

Here, $k = c/NBA = \text{galvanometer constant}$.

Thus the deflection angle of the galvanometer is proportional to the current through it.

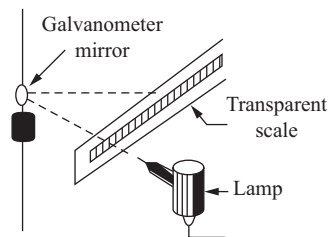


FIG. 10.26

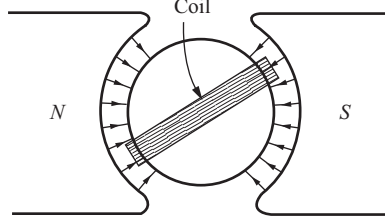


FIG. 10.27

10.11 MOVING COIL BALLISTIC GALVANOMETER

Principle. When a current is passed through a coil, suspended freely in a magnetic field, it experiences a force in a direction given by Fleming's left hand rule.

Construction. It consists of a rectangular coil of thin copper wire wound on a non-metallic frame of ivory (Fig. 10.28). It is suspended by means of a phosphor bronze wire between the poles of a powerful horse-shoe magnet. A small circular mirror is attached to the suspension wire. Lower end of the coil is connected to a hair-spring. The upper end of the suspension wire and the lower end of the spring are connected to terminals T_1 and T_2 . A cylindrical soft iron core (C) is placed symmetrically inside the coil between the magnetic poles which are also made cylindrical in shape. This iron core concentrates the magnetic field and helps in producing *radial* field.

The B.G. is used to measure *electric charge*. The charge has to pass through the coil as quickly as possible and before the coil starts moving. The coil thus gets an impulse and a throw is registered.

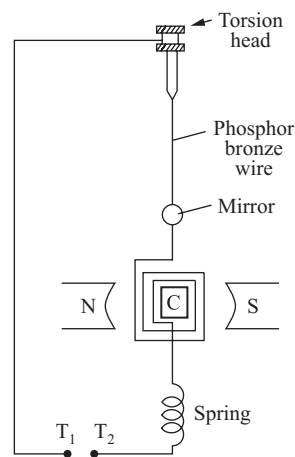


FIG. 10.28

To achieve this result, a coil of high moment of inertia is used so that the period of oscillation of the coil is fairly large. The oscillations of the coil are practically undamped.

Theory. (i) Consider a rectangular coil of N turns placed in a uniform magnetic field of magnetic induction $\mathbf{B}$ (Fig. 10.29). Let l be the length of the coil and b its breadth.

$$\text{Area of the coil} = A = lb.$$

When a current i passes through the coil,

$$\text{torque on the coil} = \tau = NiBA. \quad \dots (1)$$

If the current passes for a short interval dt , the angular impulse produced in the coil is

$$\tau dt = NiBA dt \quad \dots (2)$$

If the current passes for t seconds, the total angular impulse given to the coil is

$$\int_0^t \tau dt = NBA \int_0^t i dt = NBAq \quad \dots (3)$$

Here, $\int_0^t i dt = q$ = total charge passing through the galvanometer coil.

Let I be the moment of inertia of the coil about the axis of suspension and ω its angular velocity. Then, change in angular momentum of the coil = $I\omega$... (4)

$$\therefore I\omega = NBAq. \quad \dots (5)$$

(ii) The kinetic energy of the moving system $\frac{1}{2}I\omega^2$ is used in twisting the suspension wire through an angle θ . Let c be the restoring torque per unit twist of the suspension wire. Then, work done in twisting the suspension wire by an angle $\theta = \frac{1}{2}c\theta^2$

$$\therefore \frac{1}{2}I\omega^2 = \frac{1}{2}c\theta^2.$$

or $I\omega^2 = c\theta^2 \quad \dots (6)$

(iii) The period of oscillation of the coil is

$$T = 2\pi \sqrt{\left(\frac{I}{c}\right)} \text{ or } T^2 = \frac{4\pi^2 I}{c}$$

$$\therefore I = \frac{T^2 c}{4\pi^2} \quad \dots (7)$$

$$\text{Multiplying Eqs. (6) and (7), } I^2 \omega^2 = \frac{c^2 T^2 \theta^2}{4\pi^2}$$

$$\text{or } I\omega = \frac{cT\theta}{2\pi} \quad \dots (8)$$

$$\text{Equating (5) and (8), } NBAq = \frac{cT\theta}{2\pi}$$

$$\text{or } q = \left(\frac{T}{2\pi}\right) \left(\frac{c}{NBA}\right) \theta \quad \dots (9)$$

This gives the relation between the charge flowing and the ballistic throw θ of the galvanometer.
 $q \propto \theta$.

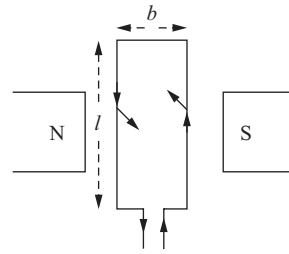


FIG. 10.29

$\left(\frac{T}{2\pi}\right)\left(\frac{c}{NBA}\right)$ is called the *ballistic reduction factor* (K).

$$\therefore q = K \theta$$

Correction for Damping in Ballistic Galvanometer

We have assumed that the whole of the kinetic energy imparted to the coil is used in twisting the suspension of the coil. In actual practice, the motion of the coil is damped by air resistance and the induced current produced in the coil. The first throw of the galvanometer is, therefore, smaller than it would have been in the absence of damping. The correct value of first throw is however obtained by applying damping correction.

Let $\theta_1, \theta_2, \theta_3, \dots$ be the successive maximum deflections from zero position to the right and left (Fig. 10.30). Then it is found that

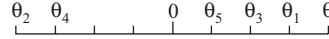


FIG. 10.30

$$\frac{\theta_1}{\theta_2} = \frac{\theta_2}{\theta_3} = \frac{\theta_3}{\theta_4} = \dots = d \quad \dots(1)$$

The constant d is called the *decrement* per half vibration.

Let $d = e^\lambda$ so that $\lambda = \log_e d$

Here λ is called the *logarithmic decrement*.

For a complete vibration,

$$\frac{\theta_1}{\theta_3} = \frac{\theta_1}{\theta_2} \times \frac{\theta_2}{\theta_3} = d^2 = e^{2\lambda}$$

Let θ be the true *first* throw in the absence of damping.

$\theta > \theta_1$, The first throw θ_1 is observed after the coil completes a quarter of vibration. In this case, the value of the decrement would be $e^{\lambda/2}$.

$$\therefore \frac{\theta}{\theta_1} = e^{\lambda/2} \approx \left(1 + \frac{\lambda}{2}\right)$$

$$\text{or} \quad \theta = \theta_1 \left[1 + \frac{\lambda}{2}\right] \quad \dots(2)$$

We can calculate λ by observing the first throw θ_1 and the eleventh throw θ_{11} .

$$\frac{\theta_1}{\theta_{11}} = \frac{\theta_1}{\theta_2} \cdot \frac{\theta_2}{\theta_3} \cdot \frac{\theta_3}{\theta_4} \cdot \frac{\theta_4}{\theta_5} \cdot \frac{\theta_5}{\theta_6} \cdot \frac{\theta_6}{\theta_7} \cdot \frac{\theta_7}{\theta_8} \cdot \frac{\theta_8}{\theta_9} \cdot \frac{\theta_9}{\theta_{10}} \cdot \frac{\theta_{10}}{\theta_{11}} = e^{10\lambda}$$

$$\text{or} \quad \lambda = \frac{1}{10} \log_e \frac{\theta_1}{\theta_{11}} = \frac{2.3026}{10} \log_{10} \frac{\theta_1}{\theta_{11}} \quad \dots(3)$$

$$\therefore q = \left(\frac{T}{2\pi}\right)\left(\frac{c}{NBA}\right)\theta_1 \left(1 + \frac{\lambda}{2}\right) \quad \dots(4)$$

Dead-beat and Ballistic galvanometers

Galvanometers are classified as (i) dead-beat or aperiodic and (ii) ballistic galvanometers.

A moving coil galvanometer in which the coil is wound on a metallic conducting frame is known as a dead-beat galvanometer. It is called “dead-beat” because it gives a steady deflection without producing any oscillation, when a steady current is passed through the coil.

Conditions for a moving coil galvanometer to be dead beat:

- (i) *Moment of inertia of the system should be small.*
- (ii) *Coil should be mounted on a conducting frame.*
- (iii) *Suspension fibre should be comparatively thicker.*

Conditions for a moving coil galvanometer to be ballistic:

- (i) The moment of inertia of moving system should be large.
- (ii) Suspension fibre should be very fine.
- (iii) Air resistance should be small.
- (iv) The damping should be small i.e., the coil should be wound on a non-conducting frame.

Differences between Dead-beat and Ballistic Galvanometers

<i>Dead-beat galvanometer</i>	<i>Ballistic galvanometer</i>
1. It measures steady current. 2. The steady deflection measures the current. 3. The coil is wound on a metal frame to increase electromagnetic damping. 4. The coil is non-oscillatory due to large damping. 5. The coil rotates due to constant torque.	1. It measures charge. 2. The throw measures the charge. 3. The coil is wound on a non-metallic frame to reduce electromagnetic damping. 4. The coil is oscillatory due to small damping. 5. The momentary passage of charge causes impulse on the coil. The torque is zero when the coil rotates.

10.12 CURRENT AND VOLTAGE SENSITIVITIES OF A MOVING-COIL GALVANOMETER

The figure of merit or current sensitivity (S_c) of a moving coil mirror galvanometer is the current that is required to produce a deflection of 1 mm on a scale kept at a distance of 1 metre from the mirror.

It is expressed in $\mu\text{A/mm}$.

The voltage sensitivity (S_v) is the p.d. that should be applied to the galvanometer to produce a deflection of 1 mm on a scale at a distance of 1 metre.

It is expressed in $\mu\text{V/mm}$.

To determine the current and voltage sensitivities of a galvanometer, the circuit shown in Fig. 10.31 is used. Two resistance boxes P and Q and a key K are connected in series with a lead accumulator of emf E . Between the ends of P , a resistance box R and the M.G., through a commutator, are connected.

A low resistance (say 1Ω) is introduced in P and a high resistance (say 9999Ω) in Q . With no resistance in R , the deflection θ of the galvanometer is found.

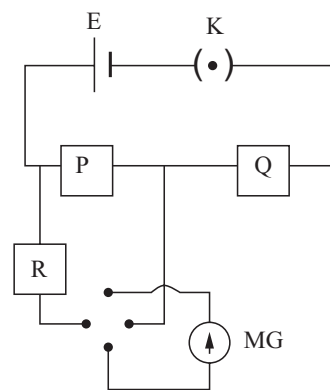


FIG. 10.31

$$\therefore \text{Voltage sensitivity, } S_v = \frac{EP}{(P+Q)\theta} \times 10^6 \mu \text{ V/mm}$$

The resistance in R is adjusted such that the deflection becomes $\theta/2$.

The resistance in R is equal to the galvanometer resistance R_g .

$$\therefore \text{Current sensitivity } S_c = \frac{EP}{(P+Q)R_g\theta} \times 10^6 \mu \text{ A/mm}$$

The experiment is repeated for various values of P keeping $P+Q$ constant.